8.8 Constant Temperature and Pressure Molecular Dynamics

The isobaric, isothermal or NPT ensemble models a system in contact with a heat bath (temperature T) and a "pressure reservoir" maintained at pressure P. The pressure is defined by the formula

$$\mathrm{P} = -\left[\frac{\partial \mathrm{E}}{\partial \mathrm{V}}\right]_{\mathrm{S,N}}$$

where E is the energy, V is volume, S is entropy. The notation $[\ldots]_{x,y}$ means that the indicated partial derivatives should be taken with respect to constant values of the indicated variables.

An isothermal-isobaric system exchanges energy with the bath in such a way as to maintain the constant temperature; at the same time the volume fluctuates to control the pressure. The state of the system can be expressed in terms of particle positions and momenta as well as the volume V, i.e. we think of the probability density as a function of $\boldsymbol{q}, \boldsymbol{p}$ and V.

Let N, V, E be the number of atoms, volume and energy, respectively, of the system of interest, and N_2, V_2, E_2 be the corresponding quantities describing the bath system; we have $\mathrm{N} + \mathrm{N}_2 = \tilde{\mathrm{N}}$, $\mathrm{V} + \mathrm{V}_2 = \tilde{\mathrm{V}}$, $\mathrm{E} + \mathrm{E}_2 = \tilde{\mathrm{E}}$ all constant. We view the entropy as a function of the other variables and expand in a Taylor series in both volume and energy:

$$\mathrm{S}(\mathrm{N}_2, \mathrm{V}_2, \mathrm{E}_2) = \mathrm{S}(\tilde{\mathrm{N}} - \mathrm{N}, \tilde{\mathrm{V}} - \mathrm{V}, \tilde{\mathrm{E}} - \mathrm{E}) \approx \mathrm{S}(\mathrm{N}_2, \tilde{\mathrm{V}}, \tilde{\mathrm{E}}) - \frac{\partial \mathrm{S}}{\partial \mathrm{E}}\mathrm{E} - \frac{\partial \mathrm{S}}{\partial \mathrm{V}}\mathrm{V}.$$

Bear in mind that here E and V are small relative perturbations of the total energy and volume. The quantity $\frac{\partial \mathrm{S}}{\partial \mathrm{E}}$ is the reciprocal temperature.

To calculate $\frac{\partial \mathrm{S}}{\partial \mathrm{V}}$ we may use *Maxwell's thermodynamic relation* [372]:

$$\left[\frac{\partial \mathrm{S}}{\partial \mathrm{E}}\right]_{\mathrm{V,N}} \left[\frac{\partial \mathrm{E}}{\partial \mathrm{V}}\right]_{\mathrm{S,N}} \left[\frac{\partial \mathrm{V}}{\partial \mathrm{S}}\right]_{\mathrm{E,N}} = -1.$$

Since $\left[\frac{\partial \mathrm{S}}{\partial \mathrm{E}}\right]_{\mathrm{V,N}}$ is T^{-1} and $\left[\frac{\partial E}{\partial \mathrm{V}}\right]_{\mathrm{S,N}}$ is $-\mathrm{P}$, we obtain

$$\left[\frac{\partial \mathrm{S}}{\partial \mathrm{V}}\right]_{\mathrm{E,N}} = \frac{1}{\left[\frac{\partial \mathrm{V}}{\partial \mathrm{S}}\right]_{\mathrm{E,N}}} = \mathrm{P}/\mathrm{T}.$$

This results in the formula

$$\mathrm{S}(\mathrm{N}_2, \mathrm{V}_2, \mathrm{E}_2) \approx \text{constant} - \frac{\mathrm{E} + \mathrm{PV}}{\mathrm{T}},$$